Topic

Library Book Search System Using Binary Search Tree (BST)

Problem Statement

Develop a C++ program to maintain library book identification numbers using a Binary Search Tree (BST). The program should allow users to insert new book identification numbers into the BST and search for a particular book ID. The BST property should be maintained, where smaller values are stored in the left subtree and larger values are stored in the right subtree. The program should display whether the searched book ID is found in the library records or not.

Algorithm

1. Start.
2. Initialize the BST root as NULL.
3. Display the menu with options to insert a book ID, search a book ID, and exit.
4. Read the user's choice.
5. If the choice is **Insert**:
    - Read the book ID.
    - If the root is NULL, create a new node.
    - If the ID is smaller than the current node's ID, move to the left subtree.
    - If the ID is greater than the current node's ID, move to the right subtree.
    - Repeat until the correct position is found.
    - Insert the book ID and maintain the BST property.
    - If the ID already exists, display a duplicate ID message.
6. If the choice is **Search**:
    - Read the book ID to be searched.
    - Compare the ID with the current node's ID.
    - If the IDs match, display "Book ID found."
    - If the ID is smaller, search the left subtree.
    - If the ID is greater, search the right subtree.
    - If the node becomes NULL, display "Book ID not found."

7. If the choice is **Exit**, terminate the program.
8. If the choice is invalid, display an error message.
9. Repeat the menu until the user chooses Exit.
10. Stop.